

MONTE CARLO SAMPLING OF A MICROCANONICAL ENSEMBLE OF CLASSICAL HARMONIC OSCILLATORS

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Two efficient methods are presented for performing exact microcanonical sampling of harmonic oscillators. It is shown that orthant sampling, which has previously been used to approximately sample microcanonical ensembles of anharmonic oscillators, is an exact sampling technique when applied to harmonic oscillators.

1. Introduction

In classical trajectory studies of intramolecular dynamics it is often necessary to accurately sample the microcanonical energy surface of a molecule. If the molecule is anharmonically coupled and there is also rotational-vibrational energy coupling, multi-dimensional selection procedures must be used since the coordinates q_i and conjugate momenta p_i do not have independent probability distributions. For anharmonic and rotationally excited triatomic molecules Bunker found that an extended rejection sampling technique could be used to accurately sample the (p, q) -space [1] ‡. However, for the larger molecule CH_3NC Bunker and Hase [3] found this procedure to be too time consuming and instead used three approximate sampling procedures which they called mode, orthant, and progressive sampling. These procedures were used ad hoc and their relationships to exact sampling methods were not discussed.

In this note we show that orthant sampling is an exact microcanonical sampling procedure for classical harmonic oscillators. We first outline an exact sampling procedure in which harmonic oscillator energies are chosen randomly and then demonstrate that orthant sampling yields the same result.

2. Method I: sampling harmonic oscillator energies

Consider a molecule consisting of n harmonic oscillators at a total constant energy $E = \sum_{i=1}^n E_i$, where the E_i are the individual harmonic oscillator energies. The probability that oscillator 1 has energy E_1 may be related to the density of states of the remaining $n - 1$ oscillators which contain energy $E - E_1$. This density is

$$N(E - E_1) = (E - E_1)^{n-2} / (n-2)! \prod_{i=2}^n h\nu_i, \quad (1)$$

If one defines $P(E_1)dE_1$ as the probability that oscillator 1 contains energy between E_1 and $(E_1 + dE_1)$, given that the total energy is E , the normalized probability that oscillator 1 contains energy E_1 is then

$$P(E_1) = (E - E_1)^{n-2} \left[\int_0^E (E - E_1)^{n-2} dE_1 \right]^{-1}. \quad (2)$$

After E_1 has been selected according to eq. (2), E_2 is chosen within the energy $E - E_1$ by the probability distribution

$$P(E_2) = (E - E_1 - E_2)^{n-3} \times \left[\int_0^{E-E_1} (E - E_1 - E_2)^{n-3} dE_2 \right]^{-1}. \quad (3)$$

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‡ For a review of the application of Monte Carlo sampling procedures to classical trajectory studies, see ref. [2].

The general expression for $P(E_i)$ is

$$P(E_i) = \left[E - \sum_{j=1}^{i-1} E_j - E_i \right]^{n-1-i} \times \left\{ \int \left[E - \sum_{j=1}^{i-1} E_j - E_i \right]^{n-1-i} dE_i \right\}^{-1}, \quad (4)$$

where the integral is evaluated between the limits 0 to $(E - \sum_{j=1}^{i-1} E_j)$. $P(E_i)dE_i$ is the probability of finding E_i in the interval $(E_i, E_i + dE_i)$ once the energies of the $i-1$ oscillators have been assigned.

The von Neumann rejection method [4] can be used to sample eq. (4). However, eq. (4) can be most conveniently sampled using the cumulative distribution function

$$C(E_i) = \int_0^{E_i} P(E_i) dE_i = R_i, \quad (5)$$

where R_i is a freshly generated random number ranging between 0 and 1. Integration of eq. (5) yields for the individual E_i

$$E_i = \left[E - \sum_{j=1}^{i-1} E_j \right] (1 - R_i^{1/(n-i)}). \quad (6)$$

Using eq. (6) to select the individual E_i , each of the harmonic oscillators in the microcanonical ensemble will have an energy distribution as given by eq. (2). This can be tested by collecting the chosen energies for each oscillator into a histogram where N_i is the number of energies in the range $(E - E_i) \pm \Delta E_i$. If eq. (2) is obeyed, plots of $\ln N_i$ versus E_i for each oscillator will be linear with a slope equal to $n - 2$. The results of such an analysis is given in table 1 for 5 oscillators containing a total energy of 100 kcal/mole. 50000 initial conditions were chosen and a ΔE of 0.5 kcal/mole was used. The plot of $\ln N_i$ versus $(E - E_i)$ for oscillator 3 is shown in fig. 1.

If coordinates q_i and conjugate momenta p_i are required for the individual oscillators they may be chosen from

$$\begin{aligned} E_i &= f_i q_i^2 / 2n + p_i^2 / 2m_i, \\ q_i &= (2E_i / f_i)^{1/2} \cos 2\pi R_i, \\ p_i &= (2m_i E_i)^{1/2} \sin 2\pi R_i. \end{aligned} \quad (7)$$

Table 1
Slopes of Monte Carlo plots of $\ln P(E_i)$ versus E_i a)

Oscillator	Method I	Method II
1	2.91 ± 0.06 b)	2.94 ± 0.05
2	2.85 ± 0.06	3.01 ± 0.05
3	3.06 ± 0.06	3.04 ± 0.06
4	3.03 ± 0.04	2.90 ± 0.08
5	3.05 ± 0.05	2.99 ± 0.06

a) 50000 initial conditions were chosen for 5 oscillators containing a total energy of 100 kcal/mole.

b) The uncertainties are for a 95% confidence interval.

[The R_i in eq. (7) is different than the one that generates E_i in eq. (6).] It is easily shown that eq. (7) gives the correct distribution function for q_i at a specific E_i :

$$P(q_i) = \frac{1}{\pi/2(2E_i/f_i - q_i^2)^{1/2}}, \quad (8)$$

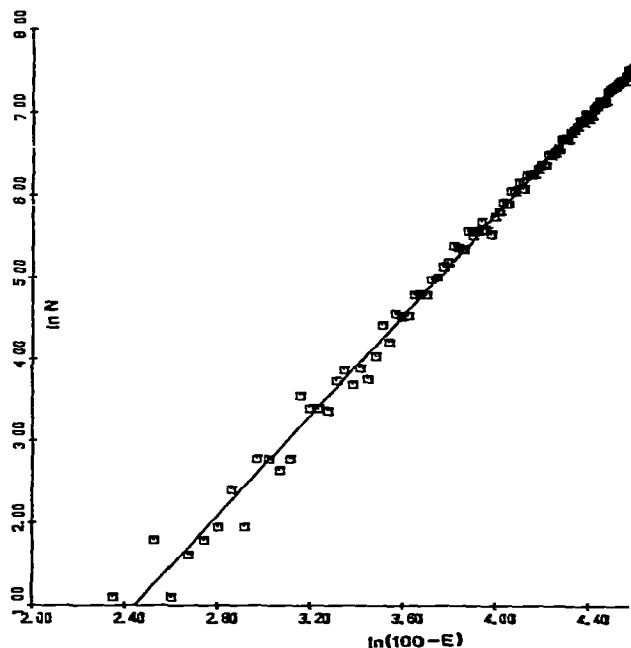


Fig. 1. Natural logarithm of the unnormalized probability distribution that oscillator 3 of a 5 oscillator system contains energy E versus natural logarithm of $(100 - E)$. 100 is the total energy of the system in kcal/mole. Sampling was performed using method I.

where

$$1.00 = \int_0^{(2E_i/f_i)^{1/2}} P(q_i) dq_i.$$

3. Method II: sampling the (p, q) -energy surface (orthant sampling)

The basic idea of orthant sampling is best illustrated if E_i in eq. (7) is written in terms of the scaled coordinates and conjugate momenta Q_i and P_i :

$$E_i = Q_i^2 + P_i^2,$$

$$Q_i = (f_i/2)^{1/2} q_i,$$

$$P_i = (1/2m_i)^{1/2} p_i. \quad (9)$$

The classical hamiltonian for a microcanonical ensemble of n harmonic oscillators is

$$H(P, Q) = \sum_{i=1}^n Q_i^2 + P_i^2. \quad (10)$$

The important point to be recognized, is that this is also the equation of a $2n$ -dimensional hypersphere of radius $H^{1/2}$. In the orthant sampling scheme proposed by Bunker and Hase [3] points are selected at random on the surface of this hypersphere by choosing a random $2n$ -dimensional unit vector and multiplying each of its components x_i by $H^{1/2}$. The randomly chosen P_i and Q_i are then

$$Q_i = H^{1/2} x_i, \quad P_i = H^{1/2} x_{n+i}, \quad (11)$$

where $\sum_{j=1}^{2n} x_j^2 = 1.00$. (In actual practice, it is inconsequential which x_i is associated with a particular Q_i or P_i .) The components x_i are chosen in accord with the relative probability distribution

$$P(x_i) = \left[1 - \sum_{j=1}^{i-1} x_j^2 - x_i^2 \right]^{(2n-i-2)/2}, \quad (12)$$

which can be conveniently sampled by the von Neumann rejection technique [4] as described by Bunker and Hase [3].

Though the above sampling scheme intuitively seems random, it is not obvious that it generates the correct harmonic oscillator and coordinate distribu-

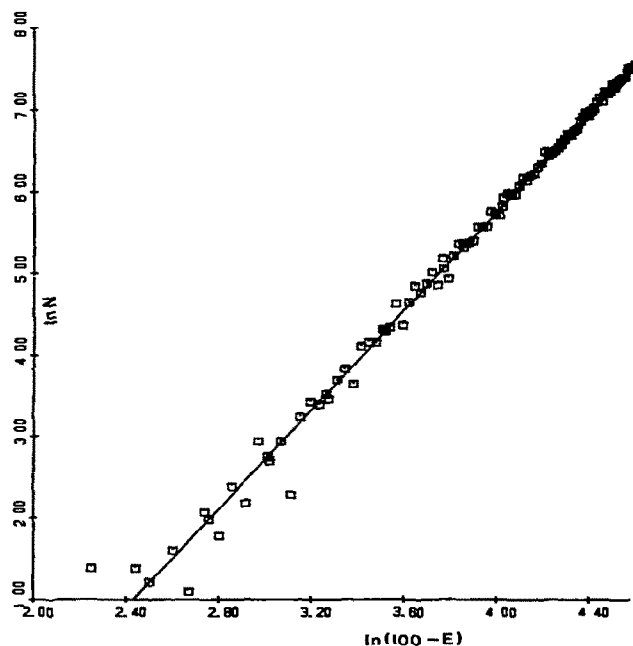


Fig. 2. Same as for fig. 1, except sampling was performed using method II (orthant sampling).

tions, as given by eqs. (2) and (8). However, that it does can be easily shown numerically. Choosing Q_i and P_i according to eq. (11), E_i values can be collected into histograms. In table 1 are listed the slopes resulting from plots of $\ln N_i$ versus $(E - E_i)$ for a 5 oscillator system at 100 kcal/mole. There is excellent agreement between these slopes and those found using method I. The plot of $\ln N_i$ versus $(E - E_i)$ for oscillator 3 is shown in fig. 2. Orthant sampling also gives the correct coordinate distribution function. This is shown in fig. 3 for oscillator 3 with energy in the range 10.0 ± 0.1 kcal/mole. The dashed line is the distribution predicted by eq. (8).

4. Conclusion

In their classical trajectory study of CH_3NC isomerization, Bunker and Hase [3] advanced orthant sampling as an approximate sampling technique for anharmonic molecules with large dimensional (p, q) -space. However, because of rotational-vibrational coupling and anharmonic potential energy terms, orthant sam-

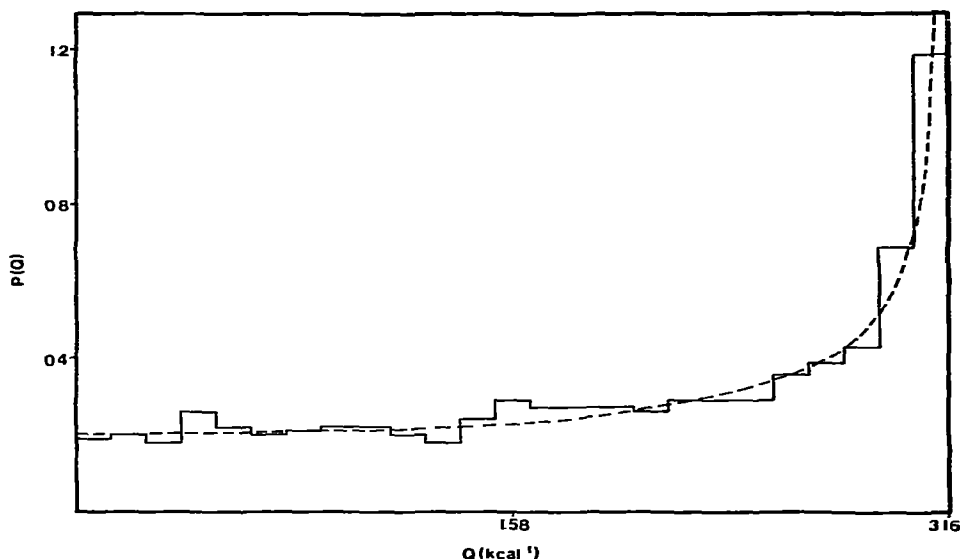


Fig. 3. Distribution function for coordinate 3 when oscillator 3 contains energy in the range 10.0 ± 0.1 kcal/mole, and method II is used for the sampling. The dashed line is given by eq. (8).

pling as outlined by Bunker and Hase does not give truly random initial conditions. For example, in the CH_3NC classical trajectory study and in a classical trajectory study of $\text{H}-\text{C}\equiv\text{C}-\text{Cl}$, $\text{H}-\text{C}\equiv\text{C}-\text{H}$, and $\text{Cl}-\text{C}\equiv\text{C}-\text{Cl}$ decomposition [5] transients appear at the $t = 0$ intercept of the calculated unimolecular life-time distributions.

In recent work a modification to the Bunker and Hase orthant sampling scheme was made so that the initial conditions give the correct relationship between total kinetic and potential energy [6]. This modification resulted in classical trajectory unimolecular life-time distributions for anharmonic $\text{H}-\text{C}-\text{C} \rightarrow \text{H} + \text{C}=\text{C}$ dissociation in excellent agreement with those found using an exact extended rejection sampling technique. This agreement suggests that it may be possible to make a simple correction to the Bunker and Hase orthant sampling scheme so that truly random initial states are chosen for anharmonic and rotationally excited hamiltonians. Such work is currently in progress.

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